

120 Days of Quant Finance

Week 3 – Linear Algebra

Day 2: Eigenvalues, Eigenvectors, and PCA Intuition

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Introduction

Eigenvalues and eigenvectors reveal the internal structure of linear transformations. In quantitative finance, they play a central role in understanding portfolio risk, covariance structure, factor models, and dimensionality reduction techniques such as Principal Component Analysis (PCA).

This section develops the mathematical foundations and intuition behind eigen decomposition, with emphasis on symmetric matrices encountered in financial applications.

Linear Transformations

A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ defines a linear transformation that maps vectors in $\mathbb{R}^n$ to other vectors in $\mathbb{R}^n$:

$$\mathbf{x} \mapsto \mathbf{Ax}.$$

In general, this transformation changes both the magnitude and direction of a vector. However, certain special vectors retain their direction under the transformation.

Eigenvectors and Eigenvalues

A nonzero vector $\mathbf{v} \in \mathbb{R}^n$ is called an **eigenvector** of $\mathbf{A}$ if:

$$\mathbf{Av} = \lambda \mathbf{v},$$

where $\lambda \in \mathbb{R}$ is the corresponding **eigenvalue**.

Eigenvectors represent invariant directions of the transformation $\mathbf{A}$, while eigenvalues quantify the scaling applied along those directions.

Geometric Interpretation

For an eigenvector $\mathbf{v}$:

- The direction of $\mathbf{v}$ remains unchanged.
- The magnitude is scaled by $|\lambda|$.
- A negative eigenvalue reverses direction.
- A zero eigenvalue collapses the vector to the origin.

Eigenvectors therefore identify principal directions along which the action of the matrix is simplest.

Symmetric Matrices

In quantitative finance, covariance and correlation matrices are symmetric:

$$\boldsymbol{\Sigma} = \boldsymbol{\Sigma}^\top.$$

Symmetric matrices possess important properties:

- All eigenvalues are real.
- Eigenvectors corresponding to distinct eigenvalues are orthogonal.
- The matrix admits an orthonormal eigenbasis.

These properties make symmetric matrices particularly well-suited for risk analysis and dimensionality reduction.

Eigen-Decomposition

A symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ can be decomposed as:

$$\mathbf{A} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top,$$

where:

- $\mathbf{Q}$ is an orthogonal matrix whose columns are eigenvectors,
- $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues.

This decomposition expresses the matrix as a rotation, followed by scaling, and then an inverse rotation.

Connection to Portfolio Risk

Let $\mathbf{\Sigma}$ denote the covariance matrix of asset returns and $\mathbf{w}$ the portfolio weight vector. Portfolio variance is given by:

$$\text{Var}(R_p) = \mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}.$$

Using eigen-decomposition:

$$\mathbf{\Sigma} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top,$$

the variance can be rewritten as:

$$\text{Var}(R_p) = (\mathbf{Q}^\top \mathbf{w})^\top \mathbf{\Lambda} (\mathbf{Q}^\top \mathbf{w}).$$

This representation shows that portfolio risk is a weighted combination of independent risk components associated with the eigenvalues of $\mathbf{\Sigma}$.

Principal Component Analysis

Principal Component Analysis (PCA) is a technique that applies eigen-decomposition to the covariance (or correlation) matrix of data.

Each principal component:

- Is an eigenvector of the covariance matrix,
- Represents an orthogonal direction of variation,
- Explains variance equal to its associated eigenvalue.

Principal components are typically ordered by decreasing eigenvalues.

Variance Explained

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ denote the eigenvalues of the covariance matrix. The proportion of total variance explained by the k -th principal component is:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}.$$

This metric determines how many components are required to capture the dominant structure in the data.

Interpretation and Limitations

PCA is a descriptive tool that identifies dominant directions of variance. However:

- Principal components are not guaranteed to be stable over time,
- High variance does not imply predictability,
- PCA does not distinguish signal from noise.

As such, PCA should be interpreted as a risk and structure analysis tool rather than a source of trading signals.

Conclusion

Eigenvalues and eigenvectors provide a powerful framework for understanding the structure of linear transformations and covariance matrices. Through PCA, these concepts enable dimensionality reduction and risk decomposition, forming a core component of modern quantitative finance.

A solid grasp of eigen-decomposition is essential for portfolio theory, factor models, and advanced machine learning methods used in financial applications.